

Residual Reinforcement Learning for Adaptive Grasp Recovery under Dynamics and Grasp Uncertainty

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CS5478 Intelligent Robots: Algorithms and Systems

Proposal Presentation

1. Problem Definition and Relevant Literature

Problem

A grasp that is initially successful may deteriorate during lifting because of uncertainty in

$$m, \quad \mu, \quad \delta p_{\text{grasp}}.$$

Possible outcomes include object slip, lateral drift or rotation, and eventual object drop.

Research question:

Can a learned recovery policy adapt corrective actions to different grasp-deterioration states?

Relevant ideas

Feedback Grasp Control

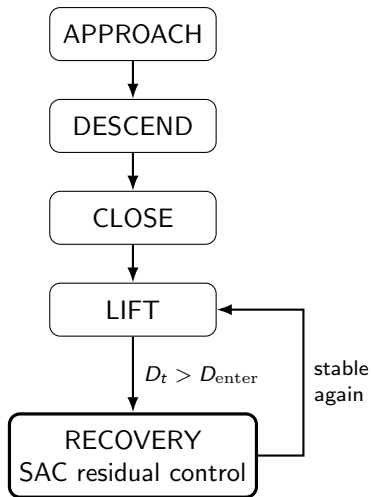
Contact feedback can be used to react to unstable grasps.

Residual Reinforcement Learning

A learned residual policy can modify an existing controller rather than replace it.

This project: mode-switched residual RL for grasp recovery during lifting.

2. Proposed Solution



Nominal operation

- ▶ Cartesian controller
- ▶ approach $\rightarrow$ grasp $\rightarrow$ lift

Deterioration detector

- ▶ relative object motion
- ▶ contact information
- ▶ ground-truth score D_t

Randomized conditions

$$m, \quad \mu, \quad \delta p_{\text{grasp}}$$

m and μ are not given directly to the recovery policy.

Recovery Action

3. Evaluation and Project Timeline

Controller Comparison

1. Nominal controller
2. Hand-designed reactive recovery
3. SAC recovery

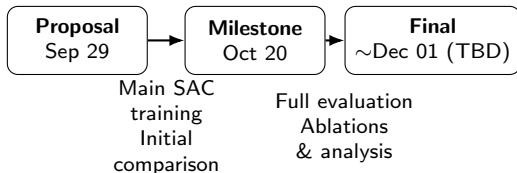
Primary Metric

$$\text{Recovery Rate} = \frac{\text{stabilized grasps}}{\text{recovery attempts}}$$

Secondary Metrics

- ▶ drop rate and relative slip
- ▶ time to recovery
- ▶ gripping effort

Timeline



Final Deliverables

Project report and final project presentation